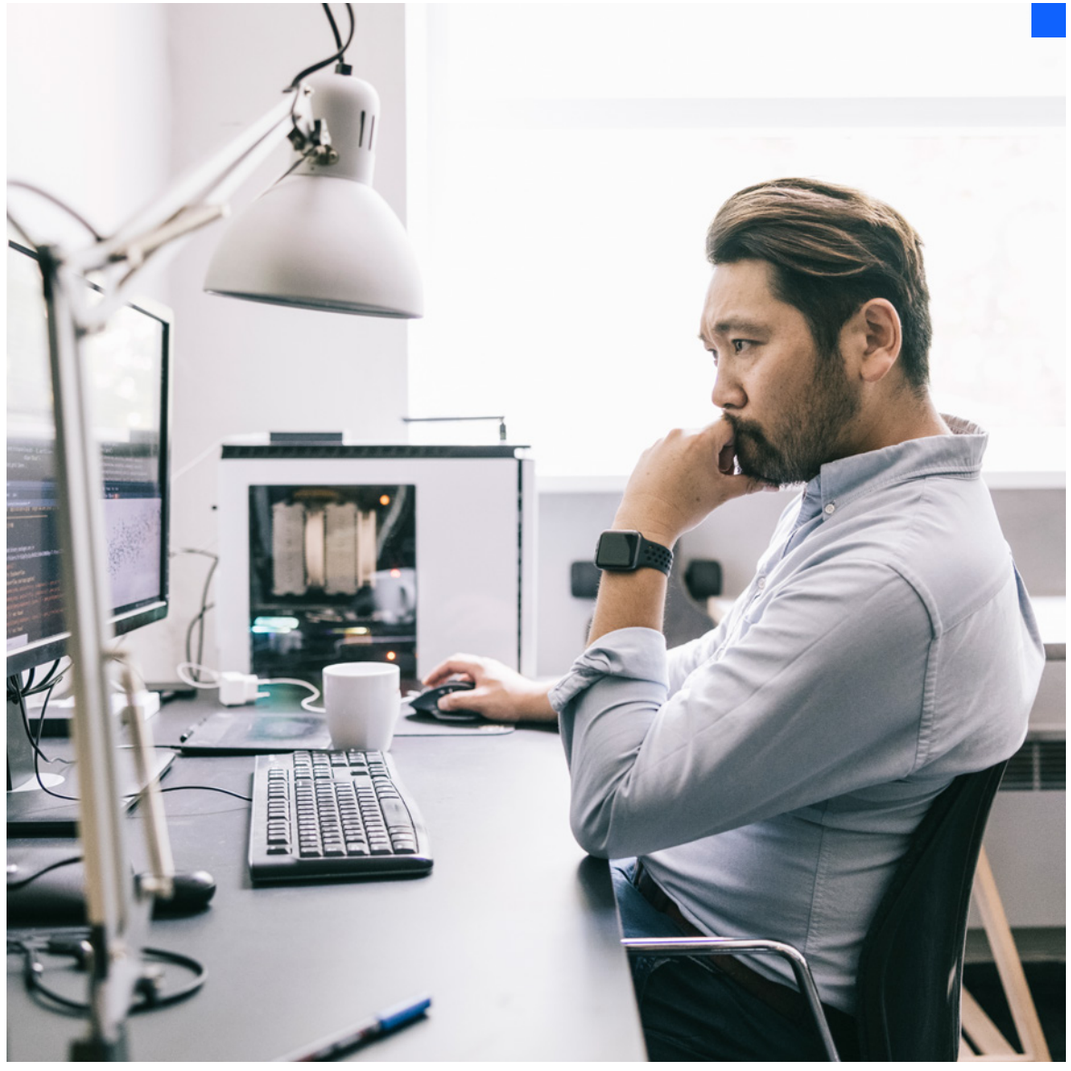
MLOps and trustworthy AI for → data-driven organizations



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01
Introduction—a data fabric approach to MLOps and trustworthy AI

People deserve AI they can trust, and leaders at every kind of organization recognize the need to adhere to ethical, explainable AI. They embrace the importance of respecting individual rights, privacy, and non-discrimination. As the outcome of AI models and insights become more business-critical, there is a dependence on them operating reliably with transparency into the process.

Trust in AI is established through the connection to the right data, coupled the automation of building, deployment and monitoring of models. Successful AI outcomes include better customer interactions, shorter time to market, and improved competitive positioning. Conversely, faulty decisions based on inaccurate data, models and processes can result in failed audits and regulatory fines. These faulty decisions can lead to loss of trust and brand reputation, higher expenses, and decreased revenues and profits for your organization.

To successfully build, deploy, and manage AI/ML models, base your method on quality data and automated data science tools and processes—an approach that requires a technology platform that can orchestrate data of many types and sources within a hybrid multicloud environment. Data fabric is an emerging technology platform that is driving better data consolidation for improved AI/ML results. The IBM team's discussion with Gartner analysts has revealed that they define data fabric similarly, as "an emerging data management design for attaining flexible, reusable and augmented data management (that is, better semantics, integration and organization of data) through metadata. Metadata drives the fabric design. Compared to traditional approaches, active metadata and semantic inference are key new aspects of a data fabric."

Data fabric provides a strong foundation for MLOps and trustworthy AI, helping to ensure that quality data can be accessed by the right people, at the right time, no matter where it resides. Data analysis is augmented with simplified, permission-based, self-service access by multiple personas (such as data scientists, analysts and business users). Users can access data for their project with confidence that the data sets are complete and accurate.

Keep reading to get the full story, or try it for yourself with our free MLOps and trustworthy AI trial.

In addition, data fabric enables multicloud data integration, data governance and compliance, and 360-degree customer intelligence (these use cases are covered in other IBM data science ebooks referenced in the conclusion of this document).

02

Why establish MLOps and trustworthy AI?

Let's look at the specific hurdles related to trustworthy AI.

Data strategy
Data collection starts with a comprehensive data strategy that's predicated on collaboration with key stakeholders including data engineering and the business. Without a strategy, data scientists waste their time hunting for quality, reliable data to address the business objective.

Data collection
It's a challenge to aggregate disparate data from internal silos, external sources, in various formats from different deployments. Algorithms and models are only as good as the data that is used to create them, and many teams lack access to the quantity and quality of data necessary to build and train accurate, unbiased AI models.

Data access, pipelines and preparation
You can introduce error and add time when you manually move high volumes of data to a target repository. Data pipelines can break, and are hard to resurrect; challenges arise with CI/CD version control, scheduling and orchestration. With weak controls for multiple personas, and the inability to track data lineage, compliance concerns rise. Manual processes can result in the access of incomplete data, the creation of biased models, and deployment delays.

Model building and deployment
Data scientists desire integrated tools to build, deploy, and train models at scale. Standalone tools can lack product documentation, which can lead to wasted time and costly errors. Informal "one-off" tools can be hard to manage and monitor, weakening governance and security. Without proper tools, it's a challenge to deploy quickly, and collaboration suffers.

Model monitoring, performance tracking and retraining
Once a model is deployed, the data scientist must continue to monitor the data and models across the entire AI pipeline. Support compliance with integrated, automated tools to track and record degradation, drift, bias, human error and historical performance. Act to ensure that deployed models are explainable; it's important that stakeholders and employees can easily understand and defend outcomes and corresponding decisions.

TechTarget reported that 83% of organizations have increased their AI budgets, and the average number of data scientists employed has grown by 76%. Yet the time required to deploy a model is also going up, with 64% of organizations taking a month or longer.[1]

03

The building blocks of MLOps and trustworthy AI

Do you want well-planned, well-executed and well-controlled AI that is built to mitigate risks and drive desired analytic outcomes? Then act to build trust and transparency across the AI lifecycle, including in the data models and processes.

Trust in data

Models and algorithms are only as good as the data that's used to create them. Trust in AI outcomes requires connection to data that is accurate, of high quality, and ready for self-service consumption by the right stakeholders. Success in AI requires a hybrid data and analytic platform with the ability to virtualize both unstructured and structured data from disparate internal and external sources. To collect and use data, you'll want to establish relevant pipelines, track lineage, and provide access to intelligent cataloging with metadata and policy management.

Trust in models

Automated, integrated data science tools backed by documentation (trials, tutorials, FAQ's) help operationalize and shorten the time to build, train and deploy AI models. The use of low/no code tools enables business users and analysts to work alongside data scientists. Automated privacy and security controls offer the right persona-based access. MLOps drives consistent and repeatable processes designed to minimize human error, increase speed to production and proactively provide AI insights to detect risk, bias and drift. The right data science tools can also increase collaboration between stakeholders including AI operations, engineering, data scientists and business analysts.

Trust in processes

For AI insights to be trusted by the project stakeholders, management and consumers, models need to be constantly managed, monitored for degradation/drift, and tweaked based on customer sensitivities around bias. Consumption of AI insights and trust in decision making starts with automated tracking of how each model is developed and deployed; this provides the insights to help identify the need to retrain or retire models. The ability to provide a transparent process and documented findings is helpful in another way: you can minimize issues around risk and regulatory compliance. Transparency and explainability in the process is crucial when you want to drive trusted AI.

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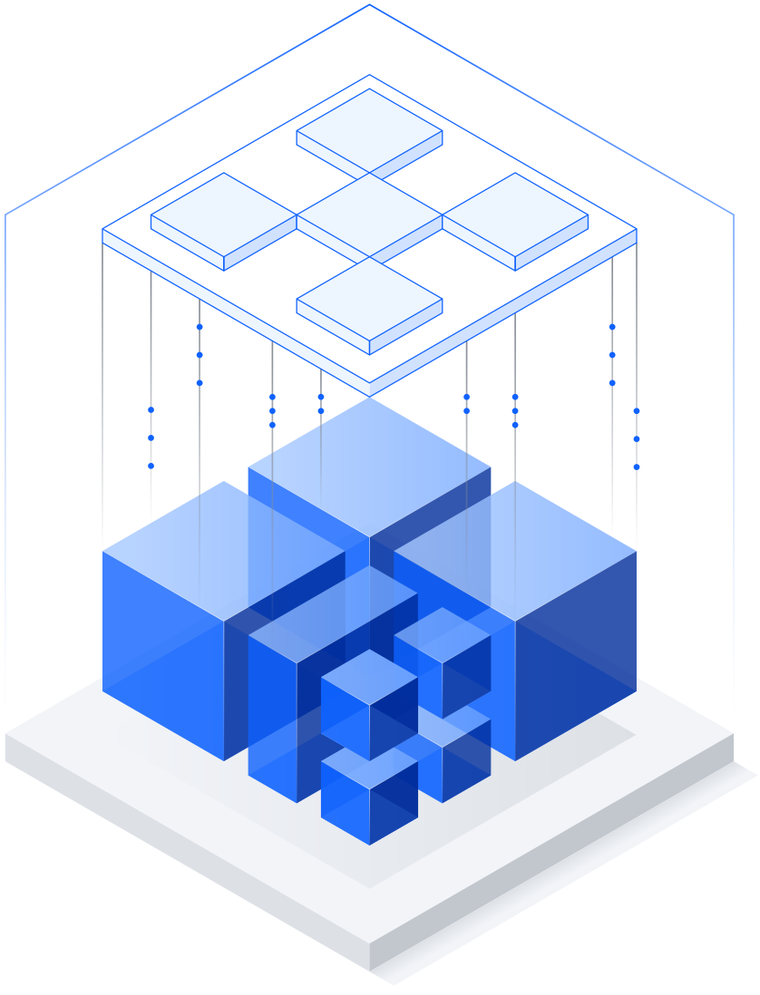
Data fabric—a holistic approach

The previous sections of this ebook describe what it takes to build trust in data, models and processes, and detail the challenges that leaders face when they don't have the right level of automation at each stage of the AI lifecycle. They also outlined the specific technological building blocks you need at each stage to establish well-planned, well-executed and well-controlled AI.

These capabilities include:

- A way to integrate data of many types and sources across diverse deployments
- Self-service access with privacy controls and a way to track lineage
- Automated model building, deployment, scaling, training and monitoring
- Automated governance to help ensure data quality and regulatory compliance



A data fabric brings multiple data sources together, augmenting rather than replacing your existing technology.

Get a more detailed view →

04

Data fabric—a holistic approach

To put all of these building blocks in place across the enterprise and to overcome the challenges of data complexity, leaders need an integrated data strategy and architecture. Enter the data fabric.

A data fabric is an architectural approach that simplifies data access in an organization to facilitate self-service data consumption. It brings together capabilities like those listed previously as part of a unified architecture, avoiding the cost and complexity that result from the integration of a plethora of point solutions. Instead of a fragmented group of products that have been stitched together, a data fabric offers a single, holistic solution that is built to work seamlessly.

A data fabric connects, governs and protects your siloed data that's distributed across a hybrid cloud landscape. It's a good way to bring the promise of your data strategy to life.

Gartner predicts that by 2023, organizations using data fabrics will dynamically connect, optimize and automate data management processes and reduce time to integrated data delivery by 30%.[2]

In the next section you'll see how other organizations have successfully implemented a data fabric approach to MLOps and trustworthy AI.

Healthcare: Penn Medicine

AI has the potential to reshape the healthcare industry by speeding up the pace of research, helping clinicians make better decisions, and improving the patient experience through personalized medicine.

Consider the case of Penn Medicine, a world-renowned academic medical center in Philadelphia whose facilities rank among the nation's top hospitals. Leaders needed a better way to predict intracranial hypertension (ICP crises) in patients with severe traumatic brain injuries (TBIs), to facilitate earlier identification and more proactive treatment of the condition.

Neurosurgeons at Penn Medicine rely on a variety of physiological time series data to make treatment decisions, but monitoring for ICP crises is a manual, labor-intensive process. The IBM Data Science and AI Elite team developed a machine learning model to predict increases in intracranial hypertension (ICP crises) in advance with a lead time of 20-30 minutes, providing enough time for neurosurgeons to intervene and potentially avoid crises. The data-driven solution made the most of machine learning by uncovering hidden patterns within the physiological time-series data. The model could provide continuous prediction in regular intervals, and if implemented in clinical workflows, could potentially improve patient outcomes after severe TBI.

Results summary
Penn Medicine was able to:

- Use predictive modeling to identify and treat intracranial hypertension earlier, to avoid further complication
- Decrease lead-time for neurosurgeons in detection and intervention
- Improve treatment decisions

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MLOps and trustworthy AI success stories

Manufacturing: ABB

Unpredictable customer demand, historic supply chain inefficiencies and labor shortages are driving the need for organizations to reevaluate manufacturing operations. Acceleration of digital technology using AI has the promise to drive operational efficiencies at scale. Productivity increases. Supply chain management improves.

ABB is a multinational corporation headquartered in Switzerland that operates in robotics, power, electrical equipment and automation technology. Leaders needed to use their historic pipe-specification data to apply advanced analytics. It was important to not just preserve and maintain the knowledge of their retiring sales engineers, but also to expand and share that knowledge.

Leaders needed an end-to-end process that could:

- Unlock their data from inadequate data formatting
- Use machine learning for analytics
- Use the output to make a real impact on their business processes

An end-to-end pipeline was created; it used advanced analytical techniques such as association mining and unsupervised machine learning to uncover hidden patterns.

The implementation of AI technologies helps traditional institutions shift to digital channels, which offer mobile and self-service options. By encapsulating the results in a prototype web application, ABB was able to envision the possible development of an internal application that could be used by its sales engineers.

Results summary
ABB was able to:

- Reduce complexity and provide transparent data infrastructure
- Reduce additional costs and resources to train new sales engineers with a new data-driven approach
- Potentially save time for the creation of pipe specification from months to days
- Run advanced analytics on historic pipe-specification data for the first time

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MLOps and trustworthy AI success stories

Financial services: NatWest Group

Today's customers expect exceptional, seamless omnichannel experiences. To create a lasting advantage and retain trust, retail banks, credit unions and other financial institutions are keen to gain insight into their customers' current and future needs—and insight into how to meet those needs.

For many people, home ownership is a dream come true. A bank-issued mortgage is often an essential ingredient in realizing that dream. But as regulations, products and processes evolve, it can be complicated to issue and obtain a mortgage. Banks are tasked with being able to access and apply accurate policy information to each customer's unique needs in real time, throughout the home buying process.

Leaders at Royal Bank of Scotland (RBS), now Natwest Group, are solving that problem with a plan for digital mortgage support. Their team created an AI-powered, cloud-based platform that empowers mortgage call center employees with real-time digital mortgage support for home buyers. The platform is called "Marge"—and is a valuable new asset to RBS' digital transformation.

Marge is intentionally personified as a member of the RBS team, and even has her own evolving personality. Marge was built directly on the cloud, embedded in RBS' existing data structures, with access to new data added every minute through content updates and customer interactions. For RBS mortgage call center employees, Marge is one single point of access for digital mortgage support for their cognitive enterprise.

While on the phone with a customer, RBS employees get quick digital mortgage support from Marge by typing keywords into a console. With cognitive enterprise technology at their fingertips, bank employees have a powerful way to support new and existing home buyers.

Results summary
Since implementing the digital mortgage support tool, RBS has seen:

- 20% improvement in customer NPS
- 10% decrease in call duration
- Better tools to empower employees as they enter the organization's digital transformation

“In the mortgage industry, change is constant. Regulation changes, product changes, process changes. It’s imperative that the customer has the support and the info they need to allow them to focus on their home buying journey.”

MaryAnn Fleming
Head of Homebuying Services
Royal Bank of Scotland

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Consider these components

IBM offers an end-to-end data science platform that provides user self-service access and integrated, supported and automated tools for model building and continuous delivery of models to production.

This set of capabilities:

- Unifies data and AI capabilities together on an integrated platform
- Increases data integration by automating engineering tasks
- Automates governance, data protection and security enabled by active metadata
- Provides collaborative tools
- Provides purpose-built AI model risk management

IBM's data fabric approach is built on a cloud-native data platform embedded with AI to help improve productivity and reduce complexity by connecting siloed data of all types, sources and workloads across a hybrid cloud environment for self-service consumption and AI models.

IBM Cloud Pak for Data
IBM Cloud Pak® for Data is a platform built specifically with a data fabric architecture in mind to predict outcomes faster and allow you to collect, organize and analyze your data, no matter where it may reside. The platform thus helps to improve productivity and reduce complexity by building a data fabric that connects siloed data distributed across a hybrid cloud landscape.

Make technology operations more agile and impactful while reducing the costs and risks of bringing AI and DevOps operations together. With its extensive ecosystem, IBM Cloud Pak for Data helps you extend your open and third-party investments and tap into IBM's latest AI innovations. As part of building your AI engineering, you can modernize your information architecture to be future-ready.

Learn more about
IBM Cloud Pak for Data →

IBM Watson Studio
IBM Watson® Studio includes tools to explore data, build models (either visually or with code), deploy and monitor these models with end-to-end lifecycle explainability and fairness. Watson Studio uses MLOps to simplify model production from any tool, provides automatic model retraining, drives transparency and monitors models over time for accuracy and bias.

As an integral feature of Watson Studio, IBM Factsheets tracks how each model is developed and deployed. Increased transparency provides information for AI consumers to better understand how a model or service was created to help determine if is appropriate to use for a specific situation or need.

Learn more about
IBM Factsheets →

Learn more about
IBM Watson Studio →

06

Consider these components

IBM Watson Knowledge Catalog
IBM Watson® Knowledge Catalog provides intelligent cataloging, with automated metadata collection and policy management to ensure the details of a model are automatically collected and stored for maximum transparency and repeatability. It ensures that models are impartial, address bias, are explainable and adapt to changing model parameters.

Learn more about
IBM Watson Knowledge Catalog →

IBM OpenPages with Watson
IBM OpenPages with Watson helps you identify, manage, monitor and run reports on risk and regulatory compliance across the entire model development lifecycle, with no changes for the requirements to the current AI/ML tools used today.

Learn more about
IBM OpenPages →

IBM also offers open-source AI governance toolkits. AI Fairness 360 helps examine, report and mitigate bias in models throughout the AI application lifecycle. AI Explainability 360 includes metrics for explaining a model's processes and decision-making.

07
Create your ideal MLOps and trustworthy AI solution

If you're eager to have even greater trust in your AI, we encourage you to take advantage of a few free IBM resources. Foremost is the MLOps and trustworthy AI trial, which will provide hands on experience with a data fabric designed for this use case. Second, review the information on our MLOps and trustworthy AI website to learn even more about the benefits. And, finally, reach out to one of our experts either by scheduling a time online, talking with your IBM representative or reaching out to one of our business partners.

Learn more about the role of a data fabric in building a data strategy and architecture with The Data Differentiator.

Check out the other three data fabric use case ebooks:

Multicloud Data Integration
Data Governance and Privacy
Customer 360

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IBM Corporation
Route 100
Somers, NY 10589

Produced in the United States of America
September 2022

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01 “Addressing 3 infrastructure issues that challenge AI adoption,” TechTarget, May 14, 2021

02 Magic Quadrant for Data Integration Tools, Gartner, March 9, 2021.